Probability basics: random variables, independence, expectation, variance and covariance

A chat model gives every possible next word a probability. What does that number mean, and why do they all add up to exactly one?

Mathematics for AI and Machine Learning | Session 13 of 18 | Probability and statistics

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Today in one view: what you will leave with

By the end of today you can



Describe a sample space, an event, a random variable and independence, and use the rules of probability on dice

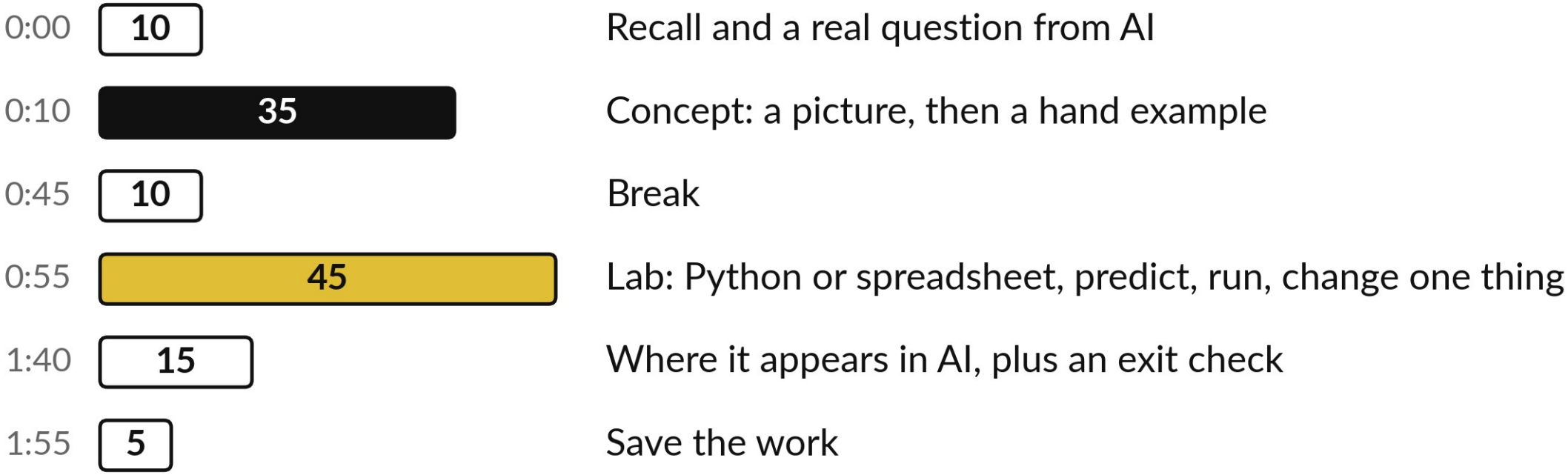


Calculate an expectation, a variance, a standard deviation, a covariance and a correlation by hand and by computer, and say whether a list is divided by n or n - 1



Read a next-word probability list from a chat model as a probability mass function

Your two hours: dice and data first, then the lab, then AI



Pictures and hand calculations first. Then the lab on dice, next-word probabilities and a small dataset.

Three questions from last time

1. Why can computing e to the power of a large score break a softmax, and what does the stable softmax do about it?

2. What does regularisation (an L2 penalty, also called weight decay) add to the loss, and why?

3. Many models use 16-bit numbers. What is the saving and what is the risk?

Answer aloud first. Then we check together.

Answers to the three questions from Session 12

Question	Short answer
Why can computing e to the power of a large score break a softmax, and what does the stable softmax do about it?	The number overflows (too large for the computer). The stable softmax subtracts the largest score first. The probabilities stay the same.
What does regularisation (an L2 penalty, also called weight decay) add to the loss, and why?	A penalty that grows with the size of the weights. It keeps weights small and reduces overfitting.
Many models use 16-bit numbers. What is the saving and what is the risk?	Less memory and less time. The largest number in the 16-bit floating-point format fp16 is only 65,504, so overflow comes sooner (bfloat16 keeps the range but loses precision).

If one answer surprised you, look back at Session 12 before the lab.

Words we use today

	Word	What it means
	**Random variable**	A rule that turns the outcome of a chance event into a number. We write it X.
	**Independent random variables**	Knowing the value of one does not change the probabilities of the other, like two separate dice.
	**Probability mass function (pmf) and density function (pdf)**	The pmf gives P(X = x) for values you can list. The pdf is a curve, and probability is the area under it.
	**Expectation, variance and standard deviation**	The probability-weighted average mu, the average squared distance from it, and its square root sigma.
	**Covariance and correlation**	How two columns move together, and the same thing scaled to lie between -1 and +1.

A sample space lists every outcome, and an event is a group of outcomes

Die 2 (rows) \ Die 1 (columns)	1	2	3	4	5	6
1	2	3	4	5	6	**7**
2	3	4	5	6	**7**	8
3	4	5	6	**7**	8	9
4	5	6	**7**	8	9	10
5	6	**7**	8	9	10	11
6	**7**	8	9	10	11	12

Two fair dice give 36 equally likely outcomes. Each cell shows the sum.

Sample space: all possible outcomes.

Event: a group of outcomes. Gold = "sum is 7".

P(sum is 7) = 6/36, about 0.167. A probability runs from 0 to 1.

The rules of probability, checked on dice

Range and total $$0 \le P(A) \le 1, \qquad P(\text{all outcomes}) = 1$$

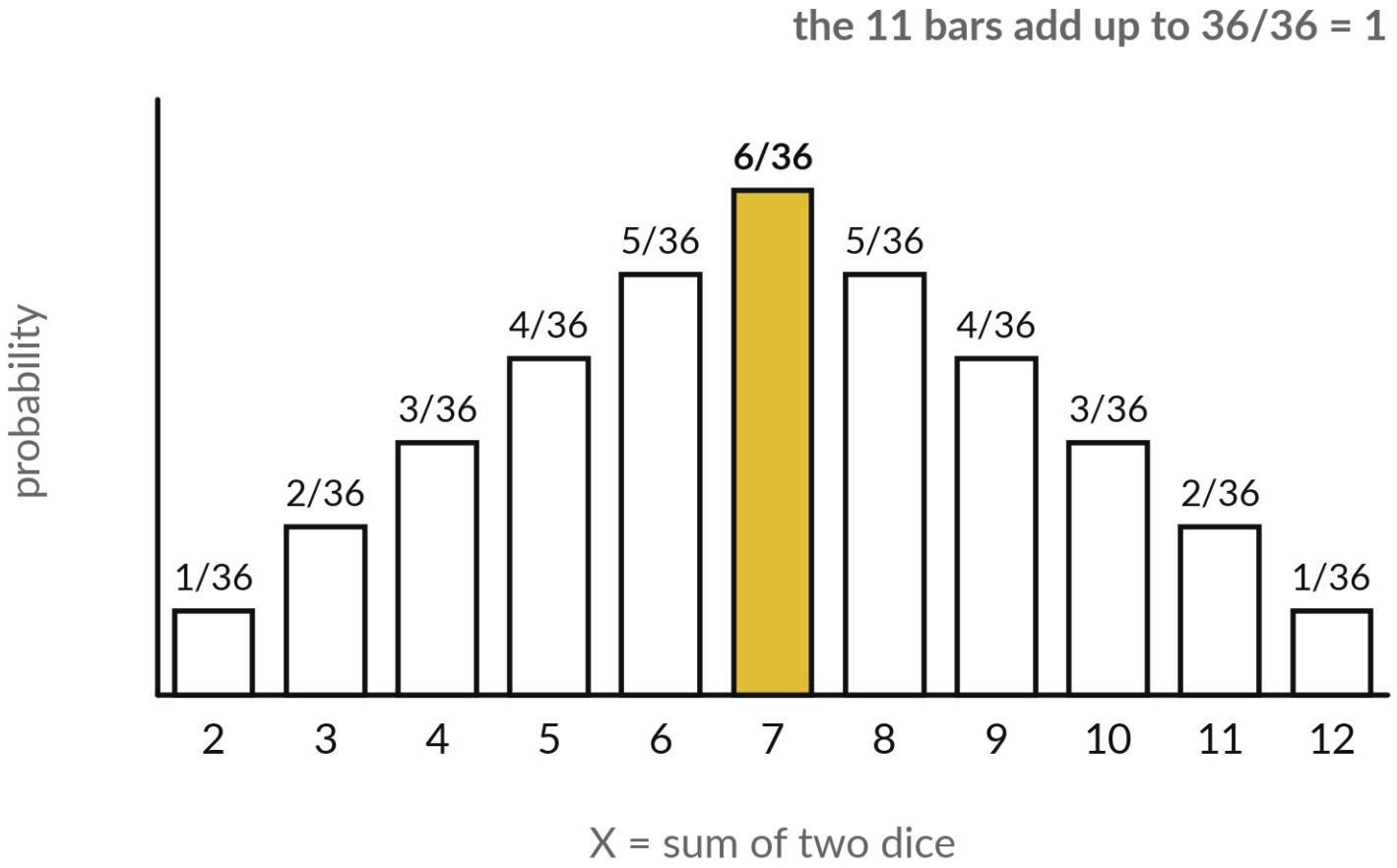
Complement $$P(\text{not } A) = 1 - P(A)$$

Either one $$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Symbol	What it means	In our data
A, B	Two events. A = the die is even {2, 4, 6}. B = the die shows more than 4 {5, 6}.	one fair die
P(A), P(B)	Their probabilities: 3/6 and 2/6.	P(A) = 0.5, P(B) = 0.333
A and B	Both at once: only the face 6.	P(A and B) = 1/6
A or B	At least one: 2, 4, 5, 6. The 6 is counted once, so we subtract it once.	3/6 + 2/6 - 1/6 = 4/6 = 0.667

Complement example: P(not even) = 1 - 3/6 = 3/6. The six faces of one die, 1/6 each, add up to 1.

A random variable turns each outcome into a number, and its pmf says how likely each number is



Let X be the sum of two dice. The bar heights are the probability mass function (pmf): p(x) = P(X = x).

A random variable X gives a number to each outcome.

P(X = 7) = 6/36 is the tallest bar (gold).

All the bars add up to 1.

Two random variables are independent when knowing one does not change the probabilities of the other

Die 2 (rows) \ Die 1 (columns)	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	**12**

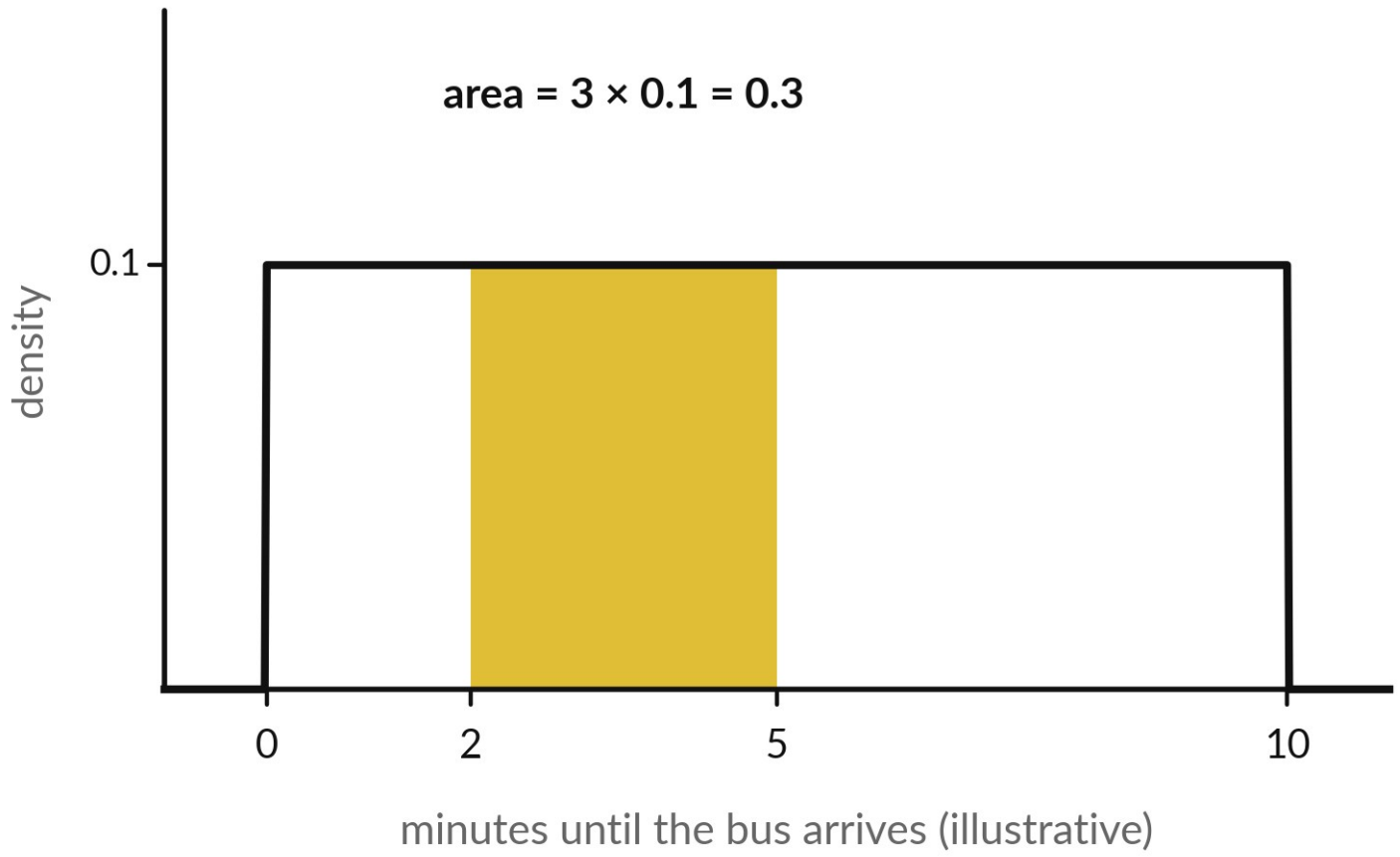
X_1 = die 1, X_2 = die 2. Independent: $P(X_1 = a \text{ and } X_2 = b) = P(X_1 = a) \times P(X_2 = b)$ for every pair a, b.

Pale: a 6 on one die. Gold: both show 6.

Gold: 1/36 = 1/6 × 1/6.

Die 1 and the sum are dependent: die 1 = 1 rules out a sum of 12.

A continuous random variable has a density curve, and probability is the area under it



A bus arrives at a random moment in a 10 minute window. This is a uniform distribution: every moment is equally likely.

Density f(x) (pdf): the curve, flat at 1/10 = 0.1, so total area = 10 × 0.1 = 1.

P(2 to 5 minutes) = width 3 × height 0.1 = 0.3.

P(exactly 3.000...) = 0. Only ranges have probability.

Probability mass function and density function, side by side

pmf (discrete X)

$$p(x) = P(X = x), \qquad \sum_x p(x) = 1$$

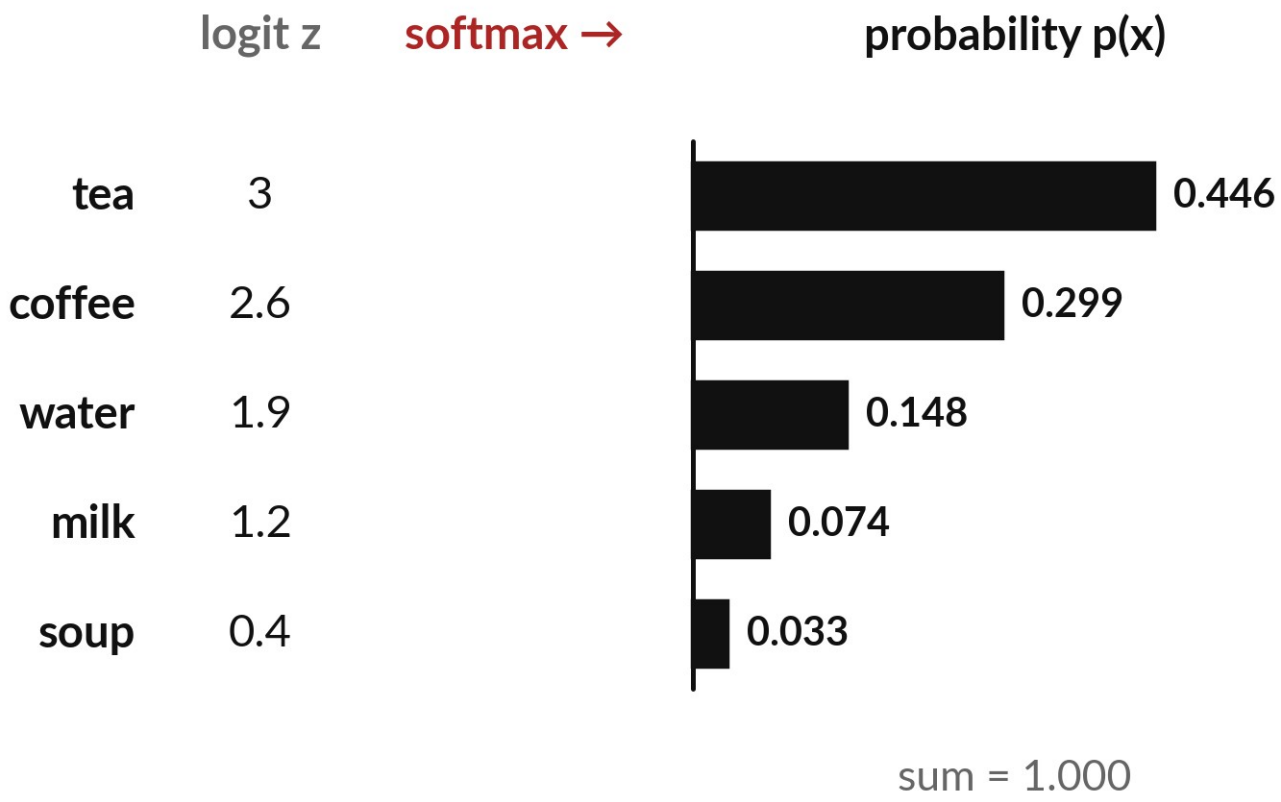
pdf (continuous X)

$$P(a \le X \le b) = \int_a^b f(x)\,dx, \qquad \int f(x)\,dx = 1$$

Symbol	What it means	In our data
p(x)	The probability that X takes exactly the value x. Never negative.	die: p(x) = 1/6
f(x)	The density: the height of the curve at x. It can be above 1.	bus: f(x) = 0.1
Σ over x	Sigma (Session 1): add over every value x that X can take.	6 faces, total 1
∫ from a to b, dx	Integral: the area under f from a to b. dx: along the x axis.	3 × 0.1 = 0.3

Both say the total probability is 1. Values you can list use a sum, a continuous curve uses an area.

A chat model gives every possible next word a probability, and the probabilities add up to 1



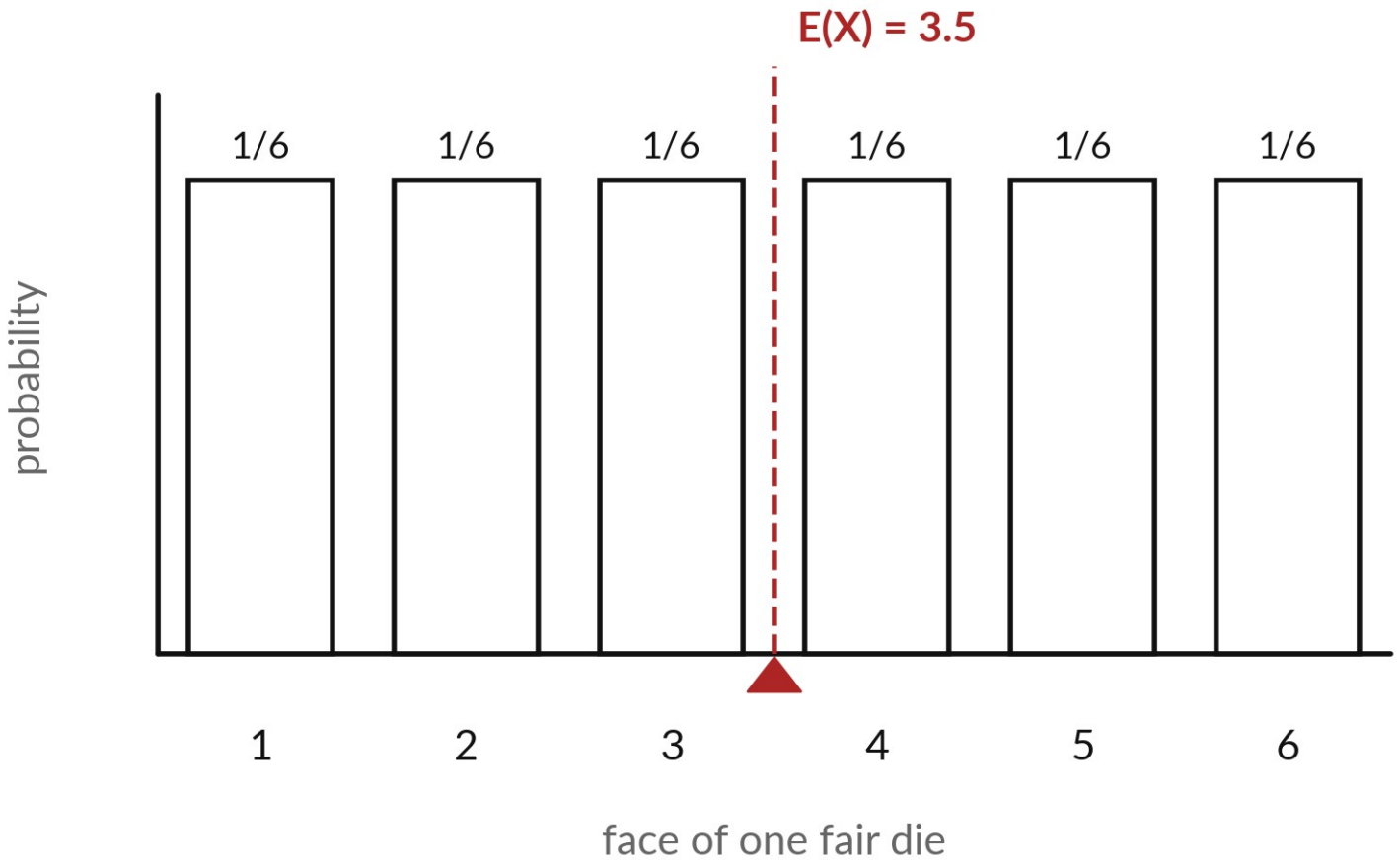
Illustrative (made-up numbers): next word after "For breakfast I drink hot ...".

Logits z: one raw score per word.

Softmax (Session 12) turns the scores into probabilities.

0.446 + 0.299 + 0.148 + 0.074 + 0.033 = 1.

Expectation is the probability-weighted average, the balance point of the bars



Expectation E(X), also called the mean and written mu, is the value the average settles on over many repeats.

Picture the bars as weights on a ruler. E(X) is where it balances.

It need not be a possible value: no die shows 3.5.

No n here: each value is weighted by its probability p(x).

Expectation: add up each value times its probability

Expectation

$$E(X) = \mu = \sum_x x\, p(x)$$

Fair die

$$E(X) = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + \cdots + 6 \cdot \tfrac{1}{6} = \tfrac{21}{6} = 3.5$$

A function of X

$$E(g(X)) = \sum_x g(x)\, p(x)$$

Symbol	What it means	In our data
E(X), mu	The expectation (mean) of the random variable X.	3.5
x and p(x)	One possible value of X, and its probability.	face 1 to 6, 1/6 each
g(x)	Any rule applied to the value, weighted by p(x).	x^2: 91/6 = 15.1667

The last line is how losses are averaged in AI: g is the loss for one outcome, E(g(X)) is the expected loss.

Expected surprise of the next word: weight each word's surprise by its probability, 1.309 nats

	A	B	C	D
	word x	**p(x)**	**surprise -ln p(x)**	**p(x) × surprise**
1	tea	0.4459	0.8077	0.3601
2	coffee	0.2989	1.2077	0.3610
3	water	0.1484	1.9077	0.2831
4	milk	0.0737	2.6077	0.1922
5	soup	0.0331	3.4077	0.1129
6		sum 1		1.3093

Surprise g(x) = -ln p(x) (ln from Session 1): a likely word surprises little, a rare word a lot. Unit: nats, because we use ln.

Tea: -ln 0.4459 = 0.8077. Soup: -ln 0.0331 = 3.4077.

E(surprise) = sum of the last column = 1.3093 nats.

Session 17 calls this average surprise the entropy, and builds the training loss from it.

Variance and standard deviation measure how far values lie from the mean

Variance

$$\mathrm{Var}(X) = \sigma^2 = \sum_x (x - \mu)^2 \, p(x)$$

Standard deviation

$$\sigma = \sqrt{\mathrm{Var}(X)}$$

Shortcut

$$\mathrm{Var}(X) = E(X^2) - \mu^2$$

Symbol	What it means	In our data
Var(X)	Variance: average squared distance from mu (units of X squared).	2.9167
sigma	Standard deviation: the square root, in the units of X.	1.7078
x - mu	How far one value lies from the expectation.	for 6: 6 - 3.5 = 2.5
$E(X^2)$	The expectation of X squared: sum of x^2 times p(x).	die: 15.1667 - 12.25 = 2.9167

We square the distances so that below-mean and above-mean values do not cancel. Again no n: the weights are p(x).

By hand: the variance of a fair die is 35/12, about 2.92

	A	B	C
	x	**x - 3.5**	**(x - 3.5) squared**
1	1	-2.5	6.25
2	2	-1.5	2.25
3	3	-0.5	0.25
4	4	+0.5	0.25
5	5	+1.5	2.25
6	6	+2.5	6.25

Each face has p(x) = 1/6, so add the last column and divide by 6.

Sum of the last column = 17.50.

Variance = 17.5 / 6 = 2.9167.

Standard deviation = square root = 1.7078: a root-mean-square distance from 3.5.

A whole population divides by n, and a sample by n - 1

Sample: divide by n - 1

Default since Session 5

np.var(x, ddof=1)

VAR.S, STDEV.S, COVARIANCE.S

Population: divide by n

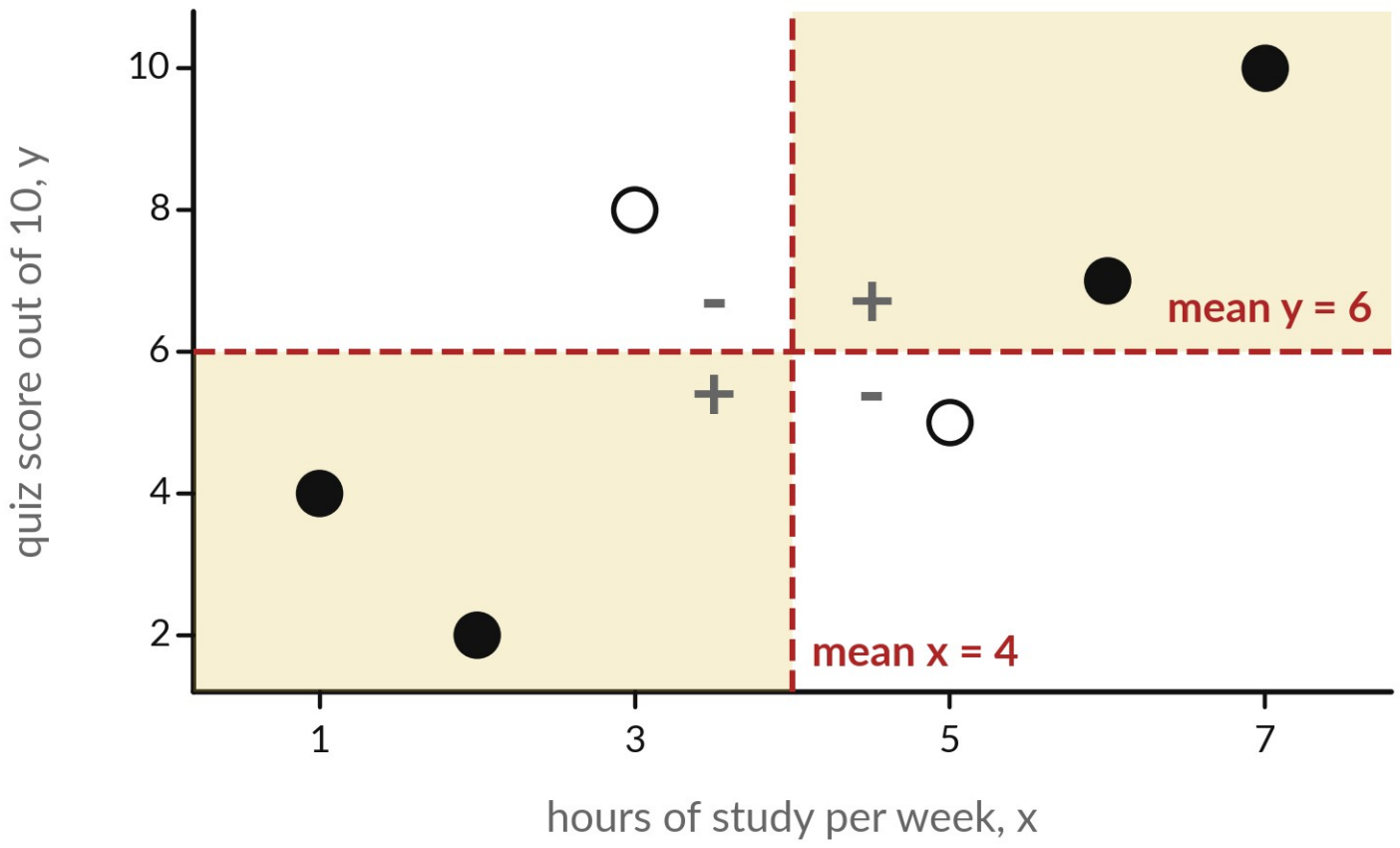
The list is all there is

np.var(x)

VAR.P, STDEV.P, COVARIANCE.P

Distances from $\bar{x}$ come out a little small; n - 1 corrects it. Var(y): 8.4 or 7.

Covariance asks whether two columns rise and fall together



Illustrative: six learners. Dashed red lines: the two means. Shaded corners: product above 0.

Dark dot: both above or both below the means. Product positive.

Open dot: product negative.

Per learner: $(x - \bar{x})(y - \bar{y})$. Add, divide by n - 1. Session 5 put it in a matrix.

By hand: the sample covariance of the six learners is 25 / 5 = 5

	A	B	C	D	E
	x	**y**	**x - 4**	**y - 6**	**product**
1	1	4	-3	-2	+6
2	2	2	-2	-4	+8
3	3	8	-1	+2	-2
4	5	5	+1	-1	-1
5	6	7	+2	+1	+2
6	7	10	+3	+4	+12
7				sum	25

The means are $\bar{x}$ = 4 hours and $\bar{y}$ = 6 points. Multiply the two distances for every learner.

Sample covariance = 25 / (6 - 1) = 5.

Sample variances: x 5.6, y 8.4.

Correlation = 5 / (2.366 × 2.898) = 0.729.

Covariance and correlation: how two columns move together

Sample covariance

$$\mathrm{Cov}(x, y) = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

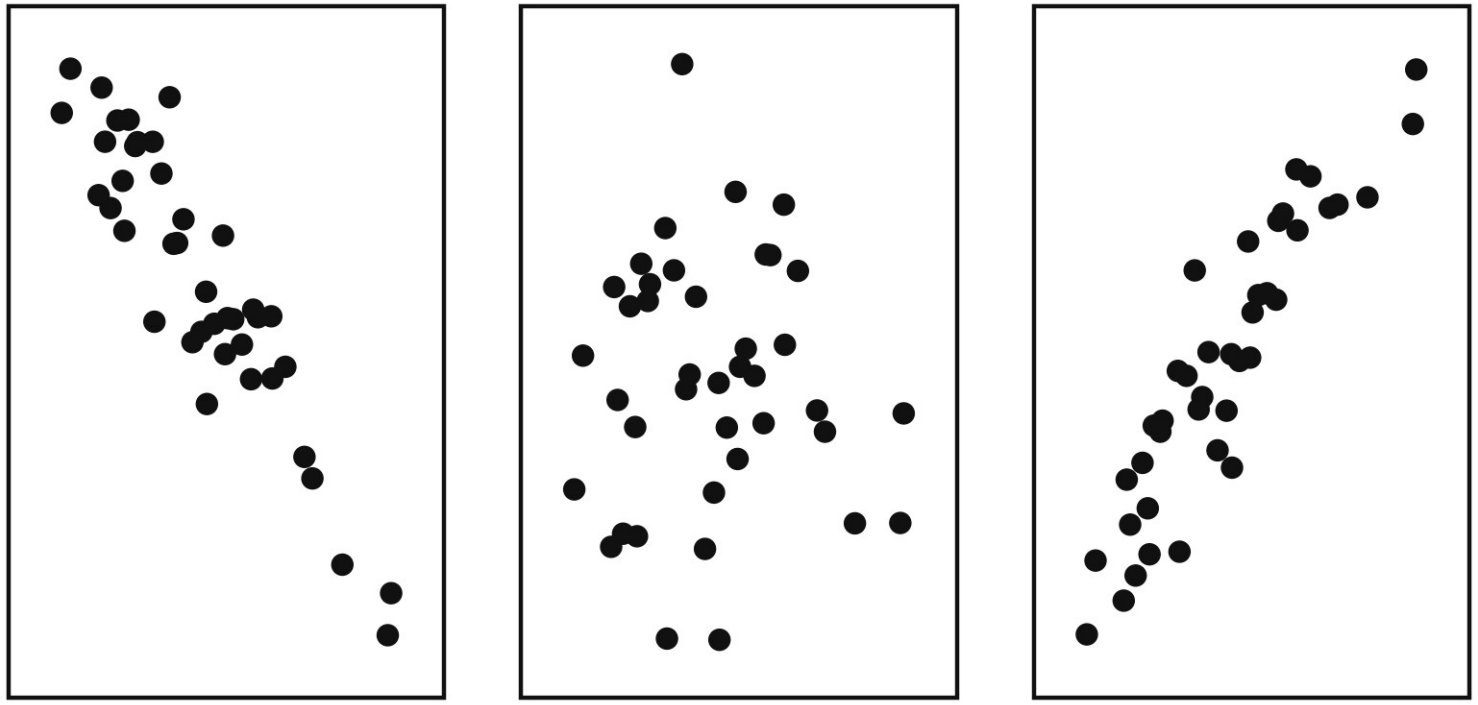
Correlation

$$r = \frac{\mathrm{Cov}(x, y)}{s_x s_y}$$

Symbol	What it means	In our data
Cov(x, y)	Sum of the products of the two distances, divided by n - 1.	5 (hours × points)
x̄, ȳ	The mean of each column.	4 and 6
s_x, s_y	The sample standard deviations (also divide by n - 1).	2.366 and 2.898
r	Covariance divided by both standard deviations. No units, between -1 and +1.	0.729

The n - 1 cancels top and bottom, so r is the same with either divisor. Cov of a column with itself is its variance.

Correlation runs from -1 to 1, and it only sees straight-line patterns



r = -0.92

falls together

r = -0.02

no pattern

r = +0.92

rises together

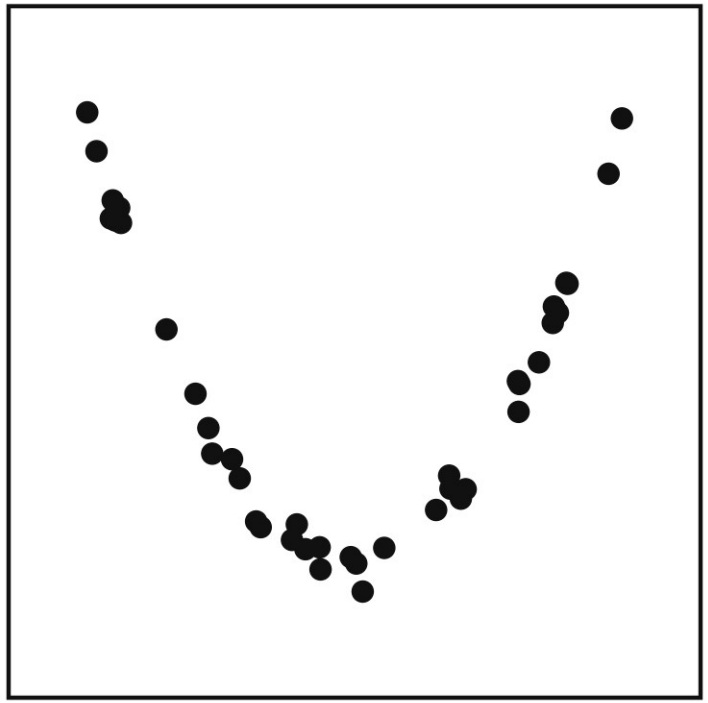
Each panel shows 40 made-up points and their correlation r.

r near +1: the columns rise together.

r near -1: one rises as the other falls.

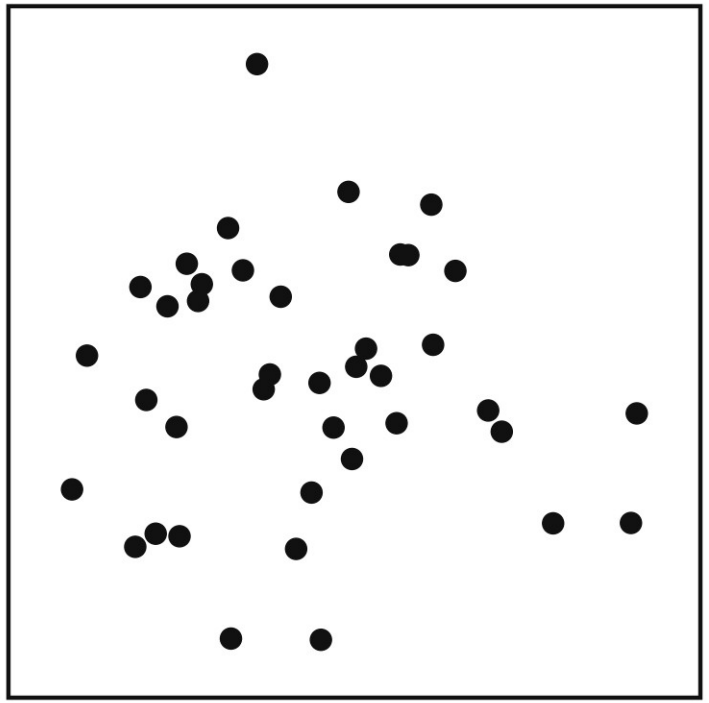
r near 0: no straight-line trend.

Common mistake: a correlation near 0 does not mean the columns are unrelated



r = -0.10

clear U shape



r = -0.02

no pattern at all

Both panels have r close to 0. Only the right one has no relationship.

Correlation measures straight-line trends only.

A strong curve can still give r near 0.

High correlation does not prove that one column causes the other.

Bridge: expectation, variance, covariance and correlation in NumPy

```
import numpy as np
x = np.array([1, 2, 3, 5, 6, 7])
y = np.array([4, 2, 8, 5, 7, 10])
np.mean(x), np.var(x, ddof=1), np.std(x, ddof=1)
np.cov(x, y)[0, 1]
np.corrcoef(x, y)[0, 1]
```

In plain words

Lines 2 and 3: the two columns as arrays.

Line 4: mean, then sample variance and standard deviation. ddof=1 divides by n - 1.

Line 5: np.cov gives a 2 by 2 matrix and divides by n - 1. Row 0, column 1 is Cov(x, y).

Line 6: np.corrcoef gives the correlation matrix. Same entry is r.

Spreadsheet: AVERAGE, VAR.S, STDEV.S, COVARIANCE.S and CORREL.

Bridge: simulate dice with a fixed seed so everyone gets the same rolls

```
rng = np.random.default_rng(13)
rolls = rng.integers(1, 7, size=60)
rolls.mean(), rolls.var(ddof=1)
rng = np.random.default_rng(13)
big = rng.integers(1, 7, size=10000)
big.mean(), big.var(ddof=1)
```

In plain words

Line 1: a random number generator. Seed 13 fixes the sequence.

Line 2: 60 whole numbers from 1 to 6 (7 is excluded).

Line 3: 60 rolls give a noisy mean (3.9667) and sample variance (3.1853).

Lines 4 to 6: start the generator again; 10,000 rolls come much closer to 3.5 and 2.9167.

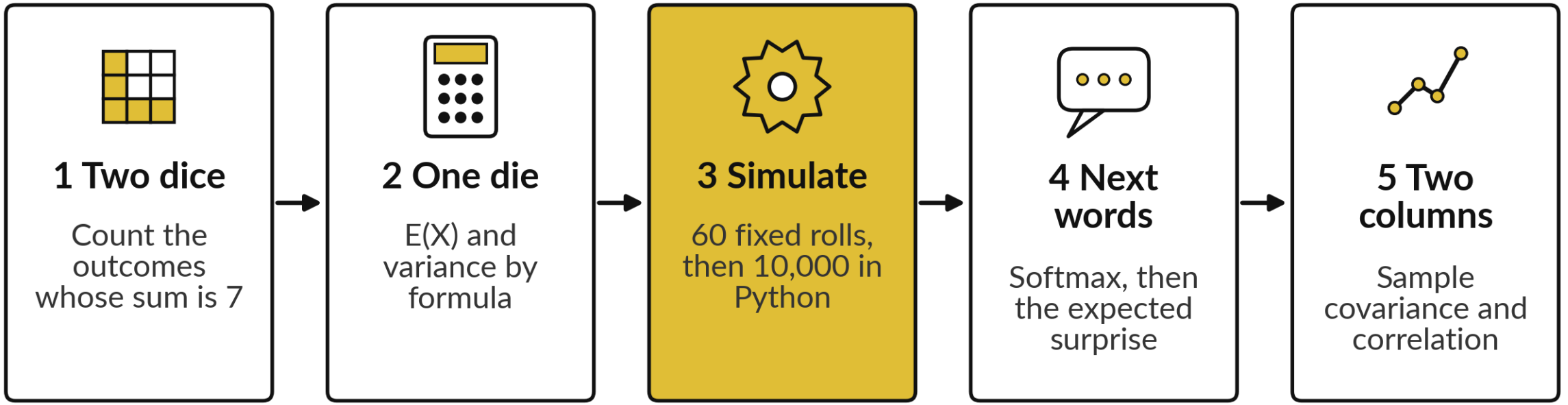
Spreadsheet: the 60 rolls are fixed values, so nothing changes.



Part 2: the lab

Close the slides. Open the notebook or the workbook.

The lab has five stages, in Python and in the spreadsheet



Python: notebook "Probability basics". Spreadsheet: workbook with the sheets Dice, Die, Words and Covariance. Predict first, run, read, then change one thing.

Check your numbers against these before you move on

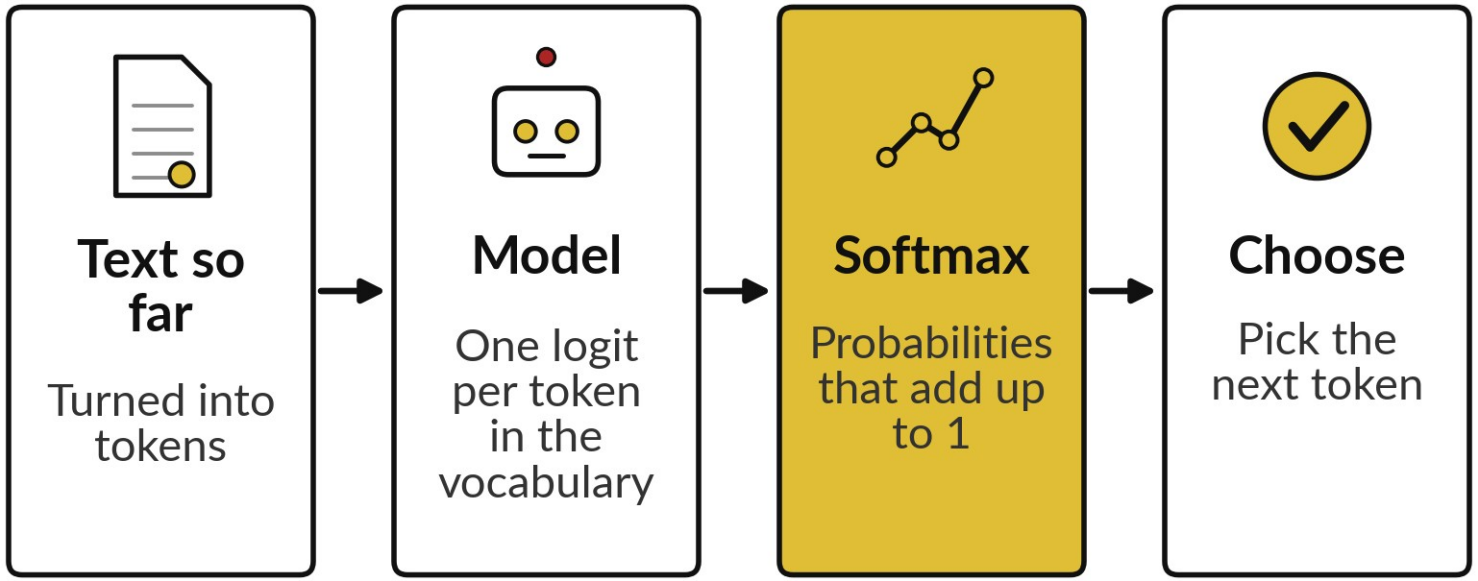
What you compute	You should see
Two dice: outcomes with sum 7, and P(sum is 7)	6 of 36, P = 0.1667
One fair die: E(X), variance, standard deviation	3.5, 2.9167, 1.7078
The 60 fixed rolls: mean and sample variance (n - 1)	3.9667 and 3.1853
Next words: P(tea), and the expected surprise in nats	0.4459 and 1.3093
x and y: sample covariance, sample variances of x and y (n - 1)	5, 5.6, 8.4
x and y: correlation	0.7290
CHANGE task: covariance dividing by n (population form)	4.1667

If your number differs, stop and find the step that went wrong. Do not change a number to match.

Errors you will meet, what they mean and what to do

	Where	What you see	What to do
✗	**Python**	NameError: name x is not defined	A cell above was not run. Use Run all from the top.
✗	**Python**	Variance of the 60 rolls 3.1322 instead of 3.1853	np.var divided by n. Add ddof=1.
✗	**Spreadsheet**	Covariance 4.1667 instead of 5	You used COVARIANCE.P. Use COVARIANCE.S for sample data.
✗	**Spreadsheet**	#NAME? in a formula	A function name is misspelled, or the sheet needs the English function names.
✓	**Both**	The 60 rolls give 3.9667, not 3.5	Correct. A small sample is noisy. Only many rolls approach 3.5.

Where this appears in AI: a language model gives every next token a probability



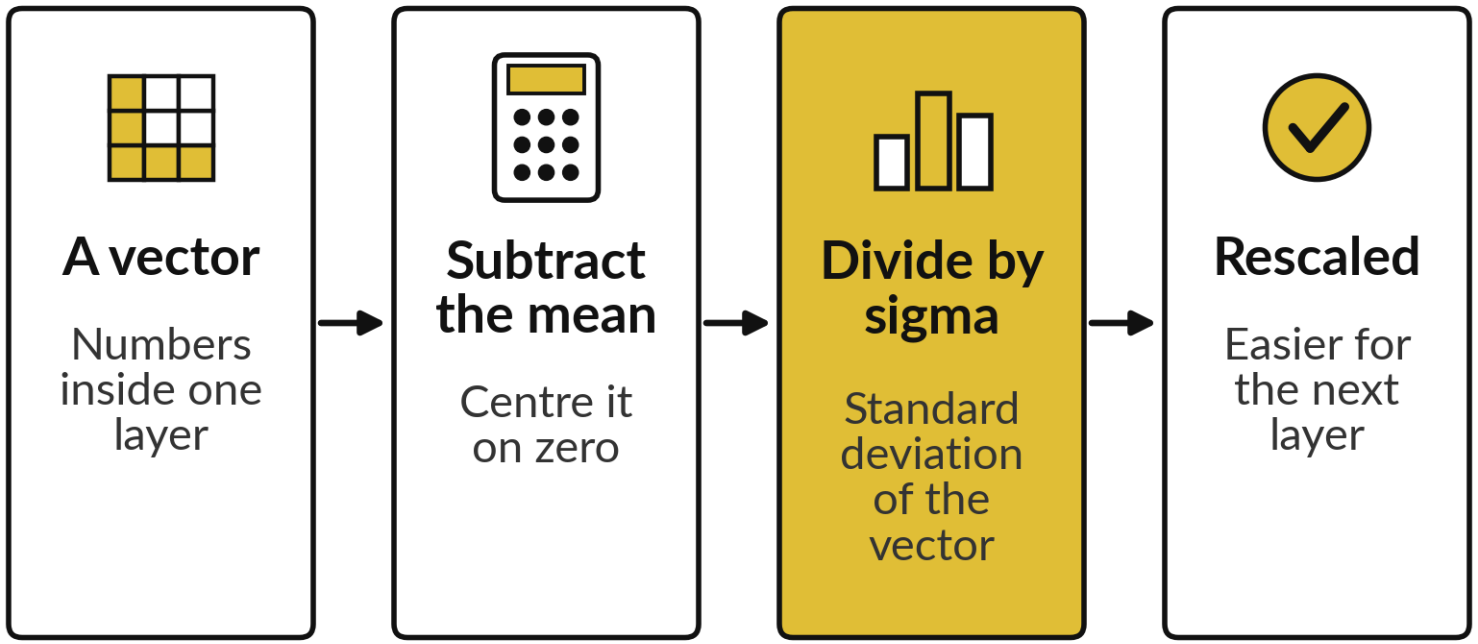
Language models of this kind output a probability mass function over the whole vocabulary.

Published transformers end in a softmax over the vocabulary.

Llama 2 (open): 32,000 tokens in the vocabulary.

Closed products do not publish details.

Where this appears in AI: mean and standard deviation rescale numbers inside the model



The original transformer uses layer normalisation: each vector is centred and scaled with its own mean and standard deviation.

Layer normalisation: Ba, Kiros and Hinton, 2016. It divides by n: the vector is the whole population.

Some newer open models use a variant (RMSNorm) that scales without centring.

Where this appears in AI: check an assistant with your numbers

8 minutes

Check an AI assistant

Ask an AI assistant for the covariance and correlation of your six pairs.

Type x = 1, 2, 3, 5, 6, 7 and y = 4, 2, 8, 5, 7, 10.

Ask for the covariance, the correlation and the divisor.

Compare with 5 and 0.729.

If it says 4.167, it divided by n.

Never paste confidential data.

You keep

A note: its value, your value, match or not.

Exit check: three questions

Question	Short answer
A model gives probabilities 0.5, 0.3 and 0.1 to three words and nothing else is possible. What is wrong?	They add to 0.9. A pmf must add to exactly 1.
Two fair dice are independent. What is P(both show 1)?	$1/6 \times 1/6 = 1/36$.
Two columns have correlation 0.02. Are they unrelated?	Not necessarily. Correlation only sees straight-line trends. Draw the scatter plot.

Save the lab and keep your five numbers

What you keep from today

Keep your saved notebook or workbook, and the numbers from the check table: E(X) = 3.5, Var(X) = 2.9167, the sample covariance 5 (n - 1), the correlation 0.729 and the next-word probabilities that add to 1.

Our question today

A chat model gives every possible next word a probability. What does that number mean, and why do they all add up to exactly one?

Next session

Session 14: distributions and sampling. We meet the binomial and the Gaussian, the central limit theorem and an A/B test.

Take away

Saved lab
Notebook or workbook

Five numbers
See the check table

Next session
Distributions and sampling

Bring
One question about chance